DeepFake Detection using RPPG Signals and XceptionNet

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Abstract. Deepfakes are AI generated videos that impersonate a person. With the advancement in technology it has become increasingly difficult to differentiate between a real and a synthetic video. Researchers have explored various methods like feature based methods and biological signals based methods to train CNN models to implement deepfake detection. PPG mapping based detection introduced in *FakeCatcher* is an effective method to detect DeepFake videos but uses a very shallow CNN model. Models like *XceptionNet* have performed well on videos but have not been tested on PPG mappings. This work uses *XceptionNet* on the already existing *FakeCatcher* architecture to comparatively analyse the accuracy of the new proposed architecture on the *FaceForensics++* dataset.

Keywords: DeepFake Detection · XceptionNet · PPG Mappings · rPPG signals.

1 Author Contributions

Student Name	Enrollment Number	Task Assigned
Anusha Garg	00701192023	write related work section, methodology, diagram and flowchart
Khushi Khorwal	02401192023	write related work section,result,conclusion and worked on initial preprocessing and training of XceptionNet.
Mishty Verma	03901192023	introduction,abstract,related work, methodology, ran existing codes and worked on XceptionNet model
Saisha Bhasin	05401192023	write dataset description and preprocessing, result and discussion, work on preprocessing data and training the CNN model

Table 1. Task Division

2 Introduction

The quality of photos and videos generated by generative models has improved significantly in recent years. It has become difficult even for professionals in Artificial Intelligence to differentiate between an AI-generated video and a real video. A deepfake video is a specific type of AI-generated video where a deep learning model is used to generate a synthetic video of a person, in which the person would be doing something that they actually never did in real life. For example, a person in a deepfake video could be speaking something which they never really spoke in real life. Good quality deepfake videos can be used in unethical ways. Due to the increased efficiency of generative models, it has become important to have a system that differentiates between real and deepfake videos [8].

Past works by different researchers have explored different methods to use deep learning or CNN models to detect deepfake videos. Various techniques like feature recognition and temporal and spatial feature analysis have been proposed in the literature. Initially, shallow CNN models like Meso-4 and MesoInception-4 proposed by Afchar et al. [1] were used, followed by deeper architectures such as XceptionNet, which was adopted as a strong baseline by Rössler et al. [9]. These models were trained on facial images and video frames to learn patterns distinguishing real and synthetic content. XceptionNet has shown strong performance on benchmark datasets such as FaceForensics++ and has often outperformed shallower CNN architectures, which is why it is widely used in deepfake detection literature [9].

Later, researchers also used spatial, temporal and other biological signals in the videos that would help classify the videos in a more generalised manner. Ciftci et al. [3] proposed FakeCatcher, one such architecture where biological g-PPG and c-PPG signals were extracted and then were analysed (1) statistically, (2) through SVM and SVR, and (3) through a shallow CNN. Their paper highlights a limitation which is the depth of their final CNN. Our paper works upon that limitation and tries to improve their pre-existing methodology by using XceptionNet as the final CNN architecture instead. As, XceptionNet is known to have performed well on purely CNN-based deepfake detection [9], and our paper would analyse if combining the biological signals with XceptionNet improves accuracy and generalisation.[3]. For the scope of this research, we will also be testing if any of the methodology have a bias based on the DeepFake category.

We will start by preprocessing the data and extracting the PPG signals from the video frames, map them and use the mapped images to train the Xcpetion-Net Model. We chose left and right cheeks and the forehead as the ROI as mentioned in FakeCatcher [2].These features will be defind using specific MediaPipe indices. Upon preprocessing and extraction, the maps will have three channels to be input into the XceptionNet model. The maps had three channels, G-PPG signals, C-PPG signal and the difference between them. For training, we will be using a downsampled version of the FaceForensics++ dataset [9] and the CelebDF-v2. We will also evaluate the model on different categories of fake data FakeForensics++ has, to evalaute any bias based on the category of fake data. The different categories of the FaceForensics++ dataset are- NeuralTextures, FaceSwap, FaceShifter, Face2Face, Deepfakes DeepFakeDetection along with original videos.

The results showed that XceptionNet performed with an accuracy and AUC score of 0.65 and 0.72 respectively, which is better than the accuracy (0.59) and AUC (0.62)of shallow models trained using the same data

Main contributions of this paper:

- To combine the perfomance of XceptionNet and rPPG signal analysis for DeepFake detection.
- Test the performance of the PPG Maps on XceptionNet to analyse and improve accuracy.
- To test baisness on the basis of the models on different categories of DeepFake videos.

3 Related Work

Table 2: Strengths and Weaknesses of Related Works

Authors	**Strengths**	**Weakness**
Local attention and long-distance interaction of rPPG for deepfake detection [3]	– The use of Remote photoplethysmography (rPPG) technology is effective in identifying deepfake detection. – The multi-scale Spatial Temporal map is used to identify heartbeat signals from facial regions. – Two stage network consisting of a mask guided local attention module and a Temporal Transformer which captures both spatial and temporal patterns in forged videos. – The proposed method (Two stage network) achieves the best result on the FF++ dataset with an accuracy of 99.38%.	– The accuracy falls against Low quality video compression (c40) to 56.96% . – The accuracy also falls for different length of video clips and does not remain consistent across different video lengths. – It also fails in head movements, lighting changes and other disturbances as heartbeat signal are easily effected by these factors and these types of images cannot be used directly to represent physiological signals. – Increasing the number of transformer layers does not help in improving the outcome.

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Table 2 – continued

Authors	**Strengths**	**Weakness**
DeepFake Detection using Xception Net [7]	– Xception net gives accuracy rates above 95% for raw and high definition videos. – Xception net performs better than shallow CNN and steganalysis techinques. Giving 80% accuracy even when videos were severely compressed. – It also outperforms human detection capabilites on NeuralTextures manipulation.	– The performance is highly dependent on face detection quality(face centeric preprocessing). – Temporal sequences and biological signals like rPPg from the videos are not during preprocessing
WildDeepfake: A Challenging Real-World Dataset for Deepfake Detection [12]	– The paper proposes two new ADDNet Architecture that used attention mask used in deepfake face fusion to improve detection accuracy. – The paper focuses on both frame level and video level detection which are 10 image level and 4 sequence level baseline networks on six different datasets. – They train 3 human annotators by check the background knowledge of deepfake generation so the unknown category handled correctly rather than mark into real/fake.	– ADDNet-3D performs worse than ADDNet-2D on dataset. – The size of dataset is very small with only 707 source videos and 7314 face sequences. – The paper proposed method archives only 76.25% accuracy on the dataset. – They used title of the collected video to labelling whether a video is real or fake as titles on the internet can be wrong and misleading.

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Table 2 – continued

Authors	**Strengths**	**Weakness**
A Comprehensive Review of Deepfake Detection Techniques Utilizing rPPG[[5]]	– The paper gives complete overview of different deepfake techniques. Early techinques(like eye movement).Then machine learning approaches like CNN and temporal analysis are discussed. Also briefs about advanced integrations like multi-modal and GANs for detection. – rPPG is non-contact technique which extract physicological signals like heart rate and blood flow patterns which deepfake models like GAN, Autoencoders, RNN are not able to replicate. – Explains rPPG extraction pipeline.The key stages of pipeline are as follow : signal capture, preprocessing , extraction and analysis. – Paper discusses various rPPG features used for deepfake detection.Some of those features are temporal patterns, pulse rate, spatial varitaions, frequency analysis. – CNNs used for deepfake detection focus on visual cues. Integrating rPPG with CNN architechture enhances the robustness.	– The physiological signals extracted are very sensitive to lightning and motion. – It is difficult to extract rPPG signals if the videos are of low qualtity or compressed. – It is a difficult task to extract these signals during real time detection. – There is no experimental comparison and qualitative analysis of different methods

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Table 2 – continued

Authors	**Strengths**	**Weakness**
FaceForensics++: Learning to Detect Manipulated Facial Images[[2]]	– The author generate 1.8 million images from 1000 real youtube videos using four different methods. – They generate three different compression quality levels of videos (raw, HQ(c23), LQ(C40)) to create realistic conditions. – To evaluate performance they conducted a study with 204 computer science study which results in a collection of 12240 human decisions. – This paper evaluates six different methods in which the XceptionNet is the best performing model achieving an accuracy of 99.26% on the raw dataset, 95.73% on HQ and 81.00% on LQ. – They use a state-of-the-art face tracking method to track the faces in the video and extract the face region from image.	– The methods required the face to be nearly front-facing to prevent the methods from degrading performances. – It includes manual screening of the clips to ensure a high quality video selection and avoid face occlusions as it includes the human bias into the dataset. – The detection approaches drops sharply for GAN based NeuralTextures approaches while future deepfakes are likely be more GAN based.
Detecting Deepfakes: A Novel Framework Employing XceptionNet-Based Convolutional Neural Networks	– The XceptionNet achieves a 96.93% accuracy and an AUC score of 97%. – The system extracts three types of facial features extraction which are eye blink patterns using Eye aspect ratio, shape feature of eye, nose and lips and deep convolutional features.	– The dataset is extremely small and imbalance which consist of total 400 movies in which 323 are fake and 77 are real. – The analysis only compares the proposed method again simple MLP-CNN and basic CNN models.

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Table 2 – continued

Authors	**Strengths**	**Weakness**
DeepFakesON-Phys: DeepFakes Detection based on Heart Rate Estimation[[4]]	– The paper archives an accuracy of 98.7% and 99.9% AUC on Celeb-DF v2 and 98.2% and 94.4 % AUC on DFDC Preview dataset. – The paper built on prior model, DeepPhys and take further with transfer learning which reduce the need to train from scratch. – The preprocessing is light as it uses face detection and normalization.	– Two different model train on two different datasets. – The paper evaluates at one frame level. – The method is sensitive to external illumination.
Celeb-DF: A Large-scale Challenging Dataset for DeepFake Forensics[[11]]	– The use of the Celeb-DF dataset achieves the highest Mask-SSIM value (0.92) among all available datasets which indicates superior visual quality. – The paper used nine detection methods on six dataset with every method tested on every dataset. – The paper introduces four improvements over the basic deepfake algorithm: higher resolution, color correction, better face masks, and reduced temporal flickering. These address weaknesses present in existing datasets.	– The dataset has an unbalanced demographic composition which comprises 88.1% Caucasians, 5.1% are Asians and 6.8% are African Americans. – Only frame level performance is detected and no temporal analysis is done. – The paper itself acknowledges there is no accepted reference free face image quality metric so they used MASK-SSIM metric.

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Table 2 – continued

Authors	Strengths	Weakness
Deepfake Detection Using XceptionNet [10]	– The paper uses the state-of-the-art XceptionNet architecture – Accuracy of the model is 91.9%, precision is 91.7% and recall is 92.2% – The paper compares XceptionNet(0.790) with multiple CNN architectures including Inception(0.782), ResNet(0.770) and VGG-16(0.715)	– The paper uses only image based information(spatial features). – The paper does not consider temporal information and physiological signals. – The dataset contains compressed videos which may affect accuracy.

DeepFake Detection using Xception Net[] has used Xception Net CNN architecture for deepfake detection. The last layer of the model is replaced by binary classifier which classifies images as real or fake. Face tracking is used crop and enlarge facial regions. The experiments are conducted on the FaceForensics++ dataset. Three different quality levels : raw, HQ(lightly compressed) , LQ(heavily compressed) are used to conduct the experiment.

In the comparative analysis conducted by Sohail and Duc [6], various architectures for deepfake detection were analyzed. The researchers evaluated Xception, Res2Net-101, EfficientNet-B7, ViT, Swin, MViT, ResNet-3D, and TimeSformer across multiple datasets to compare accuracy, log-loss, and AUC. Table 3 summarizes the performance metrics for the XceptionNet model without augmentation as reported in their study.

Table 3. Performance metrics for XceptionNet (no augmentation) across various datasets [6].

Dataset	Log-loss	AUC	Accuracy
Average (All 4 Datasets)	0.2121	96.87%	92.02%
DFDC	0.5120	91.68%	80.65%
FaceForensics++ (FF++)	0.2957	95.85%	89.03%
Celeb-DF	0.0367	99.95%	98.55%
FakeAVCeleb	0.0040	100.00%	99.85%

On the basis of the performance of these models across various datasets, with the scores for FaceForensics++ being relatively low (except for DFDC), it was concluded that this dataset is relatively challenging for models to detect or classify deepfake images. Models trained on the FaceForensics++ dataset also performed better than models trained on other datasets during inter-dataset evaluation. According to this research, datasets that are more challenging to learn provide better generalization; hence, FaceForensics++ was chosen as one of the datasets for our proposed architecture.

DeepFake Detection using Xception by Ashok V and Preetha Theresa Joy a XceptionNet model is trained to classify image as real or fake. FaceForensics++ dataset is used out of 7 categories only real and DeepFake images are used to train the model. Videos are splited into frames, each frame is used as an image. Then images are preprocessed, converted into RGB , resized, normalized and converted to tensor. Model is trained on these preprocessed images.

Detecting Deepfakes: A Novel Framework Employing XceptionNet-Based Convolutional Neural Networks by Saxena et al. proposed a deepfake detection framework that combines 68-point facial landmark extraction including eye blink detection using the Eye Aspect Ratio formula and shape feature measurements of eyes, nose, and lips with a pre-trained XceptionNet backbone to classify videos as real or fake.

WildDeepfake: A Challenging Real-World Dataset DetectionHernandez-Ortega et al. proposed DeepFakesON-Phys, a deepfake detection framework based on rPPG that uses a Convolutional Attention Network pre-trained for heart rate estimation and adapted for deepfake detection through knowledge transfer and fine-tuning.

In this paper, we are going to work upon the limitation of FakeCatcher [3] that it used a very shallow CNN (2 layer architecture only) and rather use XceptionNet to see if unsing a deeper architecture like XcpetionNet give better accuracy or generalisation.

4 Data Details

4.1 Dataset Description

Out of multiple datasets available in base paper such as FaceForensics, FaceForensics++, Celeb-DF and self collected dataset, in this project, our focus was initially on the FaceForensics++ dataset but later on we decided to include celeb-df v2 dataset as well to increase our training set. Together they provide a comprehensive and diverse collection of both real and manipulated face videos, making them well-suited for training a robust PPG-based deepfake detector.

FaceForensics++ (drive link) mainly consists of video data, both real and fake videos, where the fake videos are generated using different face manipulation techniques including DeepFakes, Face2Face, FaceSwap and NeuralTextures, which replicates realistic facial modifications in videos. The dataset is 18GB in size, consisting of 7,000 video samples (6000 deepfake, 1000 real) and 10 csv files.

CelebDF-v2(drive link) complements FF++ by introducing celebrity face videos sourced from YouTube interviews and public appearances. It contains 590 real videos and 5,639 high-quality deepfake videos. The inclusion of CelebDF-v2 introduces cross-dataset diversity and prevents the model from overfitting to the specific artifacts of a single manipulation method.

The problem is treated as a binary classification problem with two labels real and fake. The annotations for these labels are already provided by the dataset creators. The dataset also provides videos in different compression levels such as raw, high quality and low quality, which helps in analyzing the robustness of detection methods in different video conditions.

Table 4: Details of Data Attributes

Data Attribute	**Data Type**	**Brief Explanation**
Video ID	Numeric	Unique identifier assigned to each video
Video Name	Text	Name of the video file
Video Path	Text	Path to the video file in dataset directory
Frame Sequence	Image Sequence	Frames extracted from each video for analysis
Label	Categorical	Indicates whether the video is real or fake
Manipulation Type	Categorical	Type of deepfake technique used (DeepFakes, Face2Face, FaceSwap, NeuralTextures)
Compression Level	Categorical	Quality of video
Face Region (ROI)	Image	Extracted facial region from each frame
Frame Resolution	Numeric	Dimensions of video frames
Duration	Numeric	Length of the video clip

4.2 Data Pre-processing

FaceForensics++ and CelebDF-v2 datasets contain raw videos so rather than working directly with raw video frames, all videos were pre-processed into PPG (Photoplethysmography) spatial maps. Each map is a 224×224 three-channel image encoding rPPG signals extracted from different facial regions across a temporal window of the video. Real videos produce physiologically consistent PPG patterns across the face, while deepfakes produce spatially incoherent or inconsistent signals.

This signal-level representation is the core insight of out base paper and forms the foundation of all experiments in this project. Source code: Google Colab Link Preprocessed data: PPGMAPS

- Data Quality Issues: The dataset did not have the problem of null or duplicate values instead it encountered other video-data based issues.
  A major quality issue was face detection failures. In some videos, particularly the DeepFakeDetection ones, due to extreme angles, lighting, udden motion blur and partial occlusions caused the face detector to either fail entirely or produce unstable bounding boxes across frames. Thus, when a face cannot be reliably tracked, the rPPG signal cannot be extracted with spatial consistency, and the resulting PPG map becomes meaningless noise rather than a meaningful physiological representation. Such frames and their corresponding windows were excluded from the dataset entirely rather than being imputed, since any attempt to fill in missing physiological signal would introduce false biological patterns that could mislead the model.
  A secondary issue arose from temporal windowing as the PPG maps were constructed from sliding windows of video frames rather than individual frames so in some videos (particularly shorter clips in the DeepFakeDetection and CelebDF-v2 subsets) did not contain enough valid frames to generate multiple non-overlapping windows. These videos contributed only one or two maps to the dataset rather than the typical four to eight, creating a mild imbalance in per-video representation.

- Data Fusion: As mentioned above, the dataset used in this project was assembled by combining two independently sourced benchmarks, FF++ and CelebDF-v2, each with their own internal folder structure, naming conventions, and class definitions. In order to merge them, we used a unified labelling strategy since both used different terminology and FF++ separates samples into manipulation-type folders also (Deepfakes, Face2Face, etc.) under a shared under one "original" folder for real samples, while CelebDF-v2 uses a simple classification strcuture of real and fake.
  To make a unified folder, a label mapping was applied during the file-loading stage i.e. any sample originating from FF++'s original folder or CelebDF-v2's real folder was assigned label 0 (real), while all manipulation subfolders in FF++ and CelebDF-v2's fake folder were assigned label 1 (fake). This was done programmatically using `collect_paths` function that walked through each dataset's directory tree and tagged every image path with its corresponding binary label before pooling all records into a single unified list. No structural changes were made to the files themselves and thus the fusion was handled entirely at the path-label level, keeping the original data intact.

- Data Sampling: Processing all of the data from both the datasets into PPG maps and training on the full collection of around 14,000 videos would have been computationally impossible for the scope of this project. Also there was an imbalance in the ratio of real and fake videos in both the datasets, which led us to taking a sample amount of data from both.
  The dataset was pre-split at the video level into train, validation, and test partitions before PPG map generation. This splitting at the video level rather than the map level ensured that no temporal windows from the same video appear in both training and test sets, preventing data leakage.

Split	FF++ Real	FF++ Fake	CelebDF Real	CelebDF Fake	Total	Real %	Fake %
Train	6,119	5,699	4,143	3,897	19,858	51.7%	48.3%
Val	362	377	267	255	1,261	49.9%	50.1%
Test	371	325	273	260	1,229	52.4%	47.6%
Total	**6,852**	**6,401**	**4,683**	**4,412**	**22,348**	**51.6%**	**48.4%**

Table 5. Dataset split distribution of real and fake samples across FF++ and CelebDF datasets

Thus, After the full pre-processing, the combined dataset comprised of 22,348 PPG map images consisting of 11,535 real and 10,813 fake and were split across 19,858 maps for training, 1,261 for validation, and 1,229 for testing, with real and fake classes approximately balanced across all splits (roughly 52% real, 48% fake overall).

- Robustness (Augmentation): In order to improve the model's ability to generalise beyond some specific visual characteristics of training PPG maps, two forms of data augmentation were applied exclusively to the training set during the preprocessing pipeline.

It was applied during PPG map generation by processing every temporal window twice i.e. once in its original orientation and once horizontally flipped. This doubled the effective training data and encouraged the model to learn features that are not dependent on left-right facial asymmetry, which was particularly relevant for face-swap deepfakes which may introduce directional anomalies. No augmentation was applied to the validation and test sets, to ensure that evaluation metrics reflect performance on unmodified data. Roughly equal augmentation was applied across both the real and fake classes since it was label-independent and was applied uniformly to all training samples regardless of label. We deliberately chose mild augmentations so that only transformations that preserve the overall spatial structure of the map were used.

5 Methodology

The process of DeepFake detection comprised of many steps, the pipeline used is as follows facial region recognition, ROI recognition, PPG signals extraction, conversion of signal to PPG maps and feeding of these maps to XceptionNet and a two layer CNN for comparitive analysis.We will be exploring each of these steps in detail.

The facial region recognition was done using MediaPipe FaceMesh. MediaPipe FaceMesh provided 478 3-D facial landmarks which mapped to skin regions like cheeks, nose bridge, forehead, under-eye, lips, chin, and jaw.

We kept `refine_landmarks=True`, which increased the accuracy of detection of areas like eyes and lips and also improved the temporal stability by making the mesh stay fixed to the face while the person moves or speaks. This is essential for videos to keep an accurate track of the regions of interest through all frames.

To accommodate challenging lighting conditions, detection and tracking confidence thresholds were set to 0.1. This ensured that even if the video quality was poor or if the face was at a difficult angle to detect, the video was not skipped. Frames in which no faces were detected (confidence score < 0.1) were skipped. For each frame, landmark pixels were recorded for ROI computation.

Three facial ROI were defined using MediaPipe indices:

Region 1 - Left Cheek:

```
LEFT_CHEEK_ID = [116, 117, 118, 119, 100, 142, 203,
206, 207, 216, 36, 50, 93, 132,
172, 136, 150, 149, 176, 148]
```

Region 2 - Right Cheek:

```
RIGHT_CHEEK_IDX = [345, 346, 347, 348, 329, 371, 423,
426, 427, 436, 266, 280, 323, 361,
397, 365, 379, 378, 400, 377]
```

Region 3 - Mid Face:

```
MID_FACE_IDX = [
6, 197, 195, 5, 4, 1, 2,
49, 129, 203, 206, 207,
279, 358, 423, 426, 427,
```

```
116, 117, 118, 119,
345, 346, 347, 348,
0, 17, 18, 19, 94, 164,
36, 50, 93, 132,
266, 280, 323, 361,
168, 196, 174, 399,
]
```

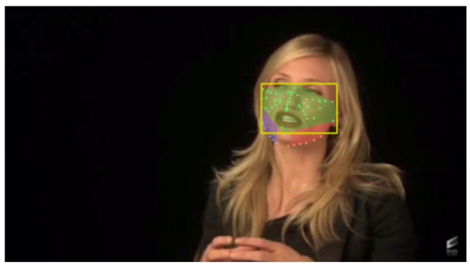

Fig. 1. Facial Region of Interest (ROI) extraction using MediaPipe FaceMesh. The face is divided into three key regions: left cheek (blue), right cheek (green), and mid-face (red).

Cheeks were chosen as ROIs because they have high density of blood flow and due to them being relatively cleaner and expressionless, better PPG signals could be extracted through them. In Deepfake videos it is possible that the blood flow of the left cheek is not coherent with that of the right cheek, hence left cheek and right cheeks were separate ROIs to analyse if any inconsistencies were present.
Regions like the forehead, lips, eyes were excluded as they did not contribute to the blood flow PPG signals we were trying to extract.
Face Preparation and ROI Processing tells how the face is prepared before signals are generated.
Initially we used Delaunay warping which transforms the face ROI into a fixed rectangle using Delaunay triangulation. However, the human face is usually shaped like an oval or diamond which creates black corners and causes loss of data as only 15 to 26 out of 32 sub-regions are included.
To overcome this, a direct bounding box approach is used. It simply crops the face ROI rather than stretching the face.
It finds 18 mid-face landmarks to create an invisible parameter. Then it creates a binary mask (white for face region and black for surroundings). Then the bounding box is applied to the entire mid-face which crops from frame and we apply mask to non-face pixels to zero value (pure black) and resize them to 128×64 rectangle. The face is divided into 32 sub-regions ($4{\times}8$ grid), each 16×16 pixels. By this all subregions become approximately equal and we can check whether the blood flow is constant across the entire face.
Some of the squares will fall on corners. To eliminate this, skin ratio validation is performed if ratio is $< 20\%$ then it uses all pixels as fallback or if ratio is $> 20\%$ then it calculate average color of only skin pixels

C-PPG (Green Channel Method): This is a subtype of remote photoplethysmography (rPPG). The code takes mean green value across 128 frames as human blood absorbs green light more effectively.
C-PPG (Chrom Method): This method includes red, green, and blue channels to create two new signals:

$$X = 3R - 2G, \quad Y = 1.5R + G - 1.5B$$

It also calculates α (ratio of standard deviations of X and Y).
Final signal:

$$Signal = X - \alpha Y$$

A Butterworth bandpass filter $[0.7, 14]$ Hz of order 3 using `scipy.signal.butter` and `filtfilt` is applied. It removes low-frequency drift and high-frequency noise, leaving only the human heart rate band.
In PPG map construction Temporal Windowing (Sliding Window) contains $\Omega = 128$ consecutive frames, stepped by 64 frames (50% overlap) for each window. These are frame 0-127, frames 64-191 and frames 128-255
Each signal matrix is converted into an image-like 2D map in which there are per sub-region normalization to [0,1] , corner infilling for indices (0,7,24,31), transpose to (128 frames $\times$ 32 regions) and scaling to unit8 after this our maps become G-PPG maps 128 $\times$ 32 and C-PPG maps 128$\times$32. Our 3 channel map become:

- Channel 1 (Red): G-PPG ,Channel 2 (Green): C-PPG,3 (Blue): $|G - C|$

The difference channel captures inconsistencies in deepfake videos.

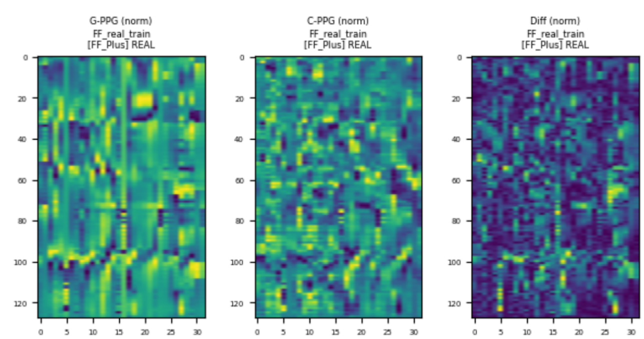

Fig. 2. Three-channel PPG map representation used as input to the CNN model. Channel 1 (Red) represents the G-PPG map, Channel 2 (Green) represents the C-PPG map, and Channel 3 (Blue) represents the absolute difference between G-PPG and C-PPG signals.

5.1 XceptionNet

Phase 1: Data Preparation The model uses 3-channel PPG maps generated from the FaceForensics++ and Celeb-DF v2 datasets.

Input: The model receives a 3-channel map consisting of G-PPG, C-PPG, and DIFF ($|G{-}PPG - C{-}PPG|$).
Target: Binary classification where 0 represents real and 1 represents fake.
To prevent overfitting, several data augmentation techniques were applied, including random resized crops, horizontal flips, brightness and contrast adjustments, shift, scale, and rotation, grid distortion, and ImageNet normalization.

Phase 2: Model Architecture The model uses a pre-trained XceptionNet backbone from the `timm` library.

Architecture:
XceptionNet Backbone (GAP) → LayerNorm(2048) → Dropout(0.4) →
Linear(2048, 512) → GELU → Dropout(0.2) → Linear(512)

XceptionNet produces a 2048-dimensional feature vector using Global Average Pooling (GAP). Layer Normalization is applied to improve training stability. Dropout is used to prevent co-adaptation of neurons. The GELU activation function introduces non-linearity. The final layer outputs predictions for binary classification (real = 0, fake = 1).
All linear layers were initialized using Xavier initialization to maintain stable gradients, with biases initialized to zero.

Phase 3: Training Strategy and Optimization During the initial epochs, all backbone parameters were frozen and only the classification head was trained. This allowed the custom head to stabilize. After the third epoch, all layers were unfrozen and the full network was fine-tuned.
The model was trained using the AdamW optimizer with decoupled weight decay to improve generalization. The learning rate helped in escaping local minima during training.
Standard cross-entropy loss was replaced with label smoothing cross-entropy to prevent overconfident predictions.
MixUp and CutMix techniques were used to generate new samples by combining different PPG maps, improving generalization.
Training was performed on a high-performance GPU. Automatic Mixed Precision (AMP) using bfloat16 was used to accelerate training and reduce VRAM usage. TF32 and cuDNN Benchmark were enabled to optimize high-throughput matrix operations on modern Tensor Cores.
The final image is saved as a PNG file.

Phase 4: Evaluation and Metrics The model was evaluated using a comprehensive metrics framework to ensure that performance was not biased by class frequency. The following metrics were used: namely Accuracy: Overall fraction of correctly classified samples, F1 Score: Harmonic mean of precision and recall, reported for the fake class specifically and as a macro average across both classes, AUROC: Area Under the Receiver Operating Characteristic curve, measuring discriminative ability across all classification thresholds, and Average Precision (AP): Area under the precision-recall curve.

Additionally, Sensitivity and Specificity were used to measure the consistency of the PPG-based detector. Sensitivity measures how accurately the system identifies fake videos, while specificity measures how accurately it classifies real videos.
The model evaluated accuracy, F1-score, and AUROC separately for each manipulation type in FaceForensics++ and Celeb-DF v2.

5.2 Baseline CNN Architecture

Phase 1: Data Preparation The model was trained on a combined dataset of FaceForensics++ and Celeb-DF v2.
The model uses 3-channel PPG maps as input, which are resized to 224×224 pixels for consistent feature extraction.
To ensure comparability with XceptionNet, the same data augmentation techniques were applied, including random horizontal flip, random vertical flip, random brightness adjustment, and random contrast adjustment.
Pixel values were normalized to the range $[0, 1]$ by dividing by 255.0.
The dataset was split as follows: with 70% Training (19,858 samples), 15% Validation (1,261 samples), and 15% Testing (1,064 samples).
The task is binary classification where 0 represents real and 1 represents fake.

Phase 2: Model Architecture The baseline model is a shallow three-block Convolutional Neural Network (CNN) trained from scratch without pretrained weights. It serves as a lower-bound reference under the same experimental conditions as XceptionNet.

Architecture:

$$\begin{gathered}\text{Input } (224, 224, 3) \rightarrow \text{Conv}(32, 3 \times 3) + \text{ReLU} + \text{MaxPool}(2 \times 2) \\ \rightarrow \text{Conv}(64, 3 \times 3) + \text{ReLU} + \text{MaxPool}(2 \times 2) \\ \rightarrow \text{Conv}(128, 3 \times 3) + \text{ReLU} + \text{MaxPool}(2 \times 2) \\ \rightarrow \text{Flatten} \rightarrow \text{Dense}(256) + \text{ReLU} + \text{Dropout}(0.5) \rightarrow \text{Dense}(2, softmax)\end{gathered}$$

Each convolutional layer uses same padding to preserve spatial dimensions before pooling.
The three convolutional blocks extract low-level spatial features (Block 1), cross-region correlations (Block 2), and higher-level temporal periodicity features (Block 3).
After flattening, the feature volume of shape $(28, 28, 128)$ is converted into 100,352 units. This is followed by a Dense(256) layer with ReLU activation and Dropout(0.5) for regularization, and a final Dense(2) softmax layer for binary classification.
The total number of trainable parameters is 25,784,130, all initialized randomly without pretrained weights.

Phase 3: Training Strategy The model follows standard deep learning training practices: namely Adam optimizer with a fixed learning rate $\eta = 10^{-3}$, early stopping where training stops if validation loss does not improve for 10 consecutive epochs with best weights restored, best model checkpoint (based on minimum validation loss) saved for final evaluation, and batch size 32 with maximum epochs 100 and random seed 42.

Phase 4: Evaluation Metrics The following metrics were used for evaluation: including Accuracy: Overall percentage of correctly classified samples, AUC-ROC: Measures the model's ability to distinguish between real and fake classes across all thresholds, and F1-Score: Harmonic mean of precision and recall, reported separately for real and fake classes, as well as macro average.

Accuracy was computed separately for each manipulation type (Original, FaceSwap, NeuralTextures, FaceShifter, Celeb-real, Celeb-fake) to identify specific weaknesses of the model.

6 Results

6.1 Experiment Setup

As discussed in Section 5 FaceForensics++ and Celeb-df datasets were used to generate physiological-maps , which were used to create three channel maps to be given as input. After preprocessing 22,351 maps of size 224x224 were generated.
Dataset Split used :
Train : 19,858 maps
Test : 1,261 maps
Validation : 1,229 maps
The training data was used to train the model parameters, testing data was used to evaluate results at the end and validation data was used for hyperparameter tunning.

For Baseline CNN model, 100 epochs were defined but the model stopped after the 40th epoch (given that the stopping patience was 10 epochs) achieving the best AUC-ROC value at 30th epoch.

Maximum of 60 epochs were defined for phase 2 training of XceptionNet. Training stopped at 28th epoch as early stopping patience was set as 12 epoch. Best value of validation AUC-ROC was achieved at 16th epoch.

Table 6. Details of machine learning algorithm parameters.

ML algo	**Parameters**
XceptionNet	Optimizer = AdamW; Learning rate (head) = 3×10^{-4}, backbone = 3×10^{-5}; Batch size = 32; Maximum epochs = 60; Dropout = 0.4; Secondary dropout = 0.2; Weight decay = 1×10^{-4}; Label smoothing = 0.1; MixUp α = 0.4; CutMix = 0.5; Early stopping patience = 12; Scheduler = CosineAnnealingWarmRestarts.
CNN	Filters = 32, 64, 128; Kernel size = 3×3; Pooling = MaxPooling2D (2×2); Dense units = 256; Dropout = 0.5; Optimizer = Adam; Learning rate = 1×10^{-3}; Batch size = 32; Maximum epochs = 100; Output layer = Softmax (2 classes).

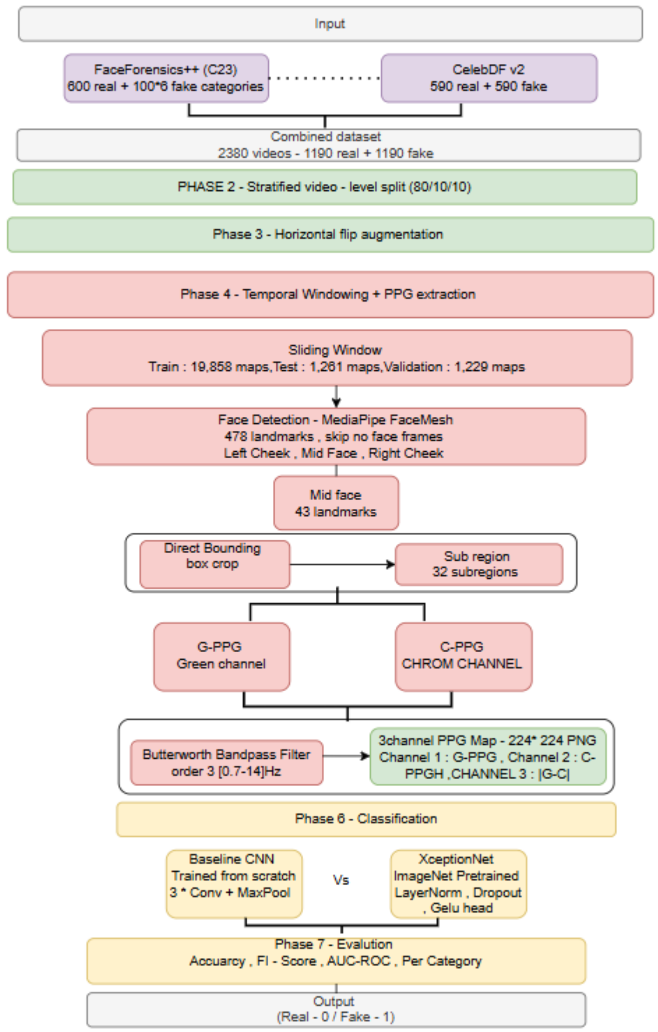

Fig. 3. Overall pipeline of the proposed deepfake detection system showing facial region extraction, ROI processing, PPG signal computation, and classification using XceptionNet , Baseline CNN.

6.2 Analysis of Results

Table 6 is side by side comparison of XceptionNet and Baseline CNN across Precision, Recall, F1-score and AUROC score. XceptionNet outperforms baseline CNN on all metrics. XceptionNet has AUROC socre of 0.72 whereas basline CNN AUROC score is 0.62, making XceptionNet classifications better.

Table 7. Class-wise Performance Comparison of Classification Models.

<table>
<tr><th>Algorithm</th><th>Class</th><th>Prec</th><th>Rec</th><th>F1</th><th>AUROC</th></tr>
<tr><td rowspan="3">XceptionNet</td><td>Real</td><td>0.73</td><td>0.54</td><td>0.62</td><td rowspan="3">0.72</td></tr>
<tr><td>Fake</td><td>0.61</td><td>0.78</td><td>0.68</td></tr>
<tr><td>Overall Accuracy</td><td colspan="3">0.65</td></tr>
<tr><td rowspan="3">Baseline CNN</td><td>Real</td><td>0.68</td><td>0.61</td><td>0.64</td><td rowspan="3">0.62</td></tr>
<tr><td>Fake</td><td>0.49</td><td>0.56</td><td>0.52</td></tr>
<tr><td>Overall Accuracy</td><td colspan="3">0.59</td></tr>
</table>

Table 7 show category wise accuracy value of XceptionNet for FaceForensics++ and Celeb-Df datasets. Deepfakes(0.94) and DeepFakeDetections(0.92) have the highest accuracy indicating that PPG maps generated on older GAN models produce stronger inconsistencies making it easier for XceptionNet to detect these correctly. Since the real videos generate clean and undistorted maps showing natural pphysiological signals but models has been trained on more diversified fake patterns it also missclassify them as fakes.

Table 8. Per-category accuracy of XceptionNet on the test set of FaceForensics++.

Dataset	**Category**	**Maps**	**Accuracy**
FF_Plus	original	371	0.53
FF_Plus	NeuralTextures	44	0.81
FF_Plus	FaceSwap	45	0.71
FF_Plus	FaceShifter	71	0.90
FF_Plus	Face2Face	58	0.89
FF_Plus	Deepfakes	55	0.94
FF_Plus	DeepFakeDetection	52	0.92
CelebDF_v2	real	273	0.54
CelebDF_v2	fake	260	0.65

Table 9. Per-category accuracy of CNN Model on the test set of FaceForensics++.

Dataset	Category	Maps	Accuracy
FF_Plus	original	371	0.62
FF_Plus	NeuralTextures	44	0.47
FF_Plus	FaceSwap	45	0.57
FF_Plus	FaceShifter	71	0.69
CelebDF_v2	real	273	0.62
CelebDF_v2	fake	260	0.54

Table 8 shows per category accuracy values of baseline CNN on FaceForensic++ and Celeb-Df dataests. Unlike XceptionNet, CNN shows no pattern in accuracy values. In comparsion to Xception it works better for real videos PPG maps but at the same it performs poorly for fake detection across all categories , confirming that it is too simple to learn patterns from PPG maps.

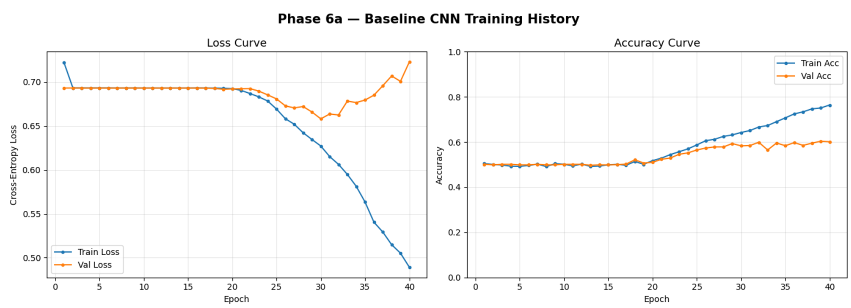

Fig. 4. Training and validation loss (left) and accuracy (right) curves of Baseline CNN model across 40 epochs

In Fig.4 for the first 6 epochs, the model showed that the accuracy of the training of the learning signal hovered between 48.9% and 50.6%, and both loss of training and validation remained near 0.693, which is exactly log(2) and the expected output of an untrained binary classifier that outputs near-uniform probabilities. This flat plateau shows that the model was stuck and gradients were insufficient to push it out of the random initialisation curve.

At 6th epoch, ReduceLROnPlateau halved the learning rate from 1e-3 to 5e-4 after five epochs due to this reduction, a very marginal improvement appeared at epoch 7, but the model remained largely stagnant through epoch 11, where a second LR halving brought it to 2.5e-4. It was after this second reduction that the model began learning meaningfully and the training loss started descending noticeably, and accuracy began increasing. The model then trained productively for approximately 29 more epochs before early stopping patience terminated training at epoch 40, at which point the best validation weights were restored.

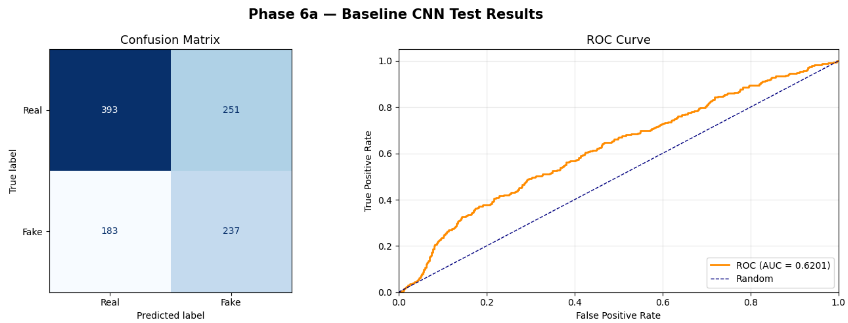

Fig. 5. Confusion matrix (left) and ROC curve (right) for the Baseline CNN model evaluated on 1,064 test PPG maps.

In Fig.5 The confusion matrix reveals 393 correctly identified real samples and 237 correctly identified fake samples, with 251 real maps misclassified as fake and 183 fake maps misclassified as real. The ROC curve achieves an AUC of 0.6201, confirming that the model's predicted probabilities carry meaningful discriminative information above the random baseline (dashed line, AUC = 0.5), although the curve's shape indicates limited sensitivity at low false positive rates , the model cannot reliably identify fakes without accepting a substantial number of false alarms.

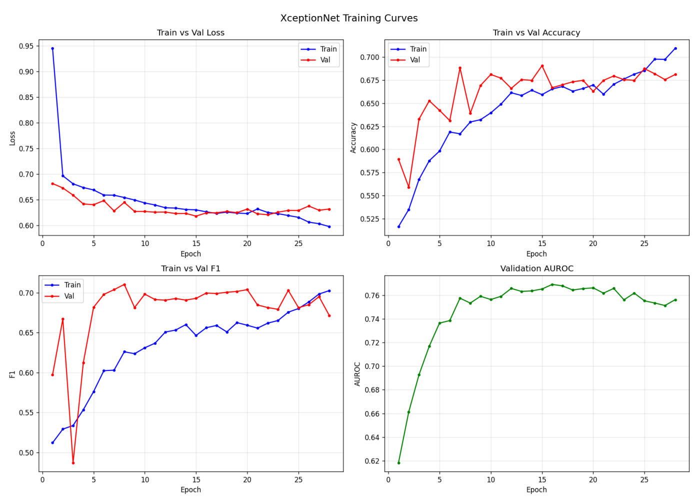

Fig. 6. Training curves of XceptionNet showing loss, accuracy, F1-score, and validation AUROC.

In Fig.6 four subplots shows epochwise progression of Train Vs Val Loss, Train Vs Val F1, Train Vs Val Accuracy and Validation over 27 epoch. **Loss** : train loss steeply decreases and then continuse to decrease towards 0.40, whereas val

loss stops changing around 0.65 indicating overfitting. **Accuracy** : Both train and val curve rises initially but train curve surpasses val curve. Val curve accuracy is in range of 65-68% showing that the model is not generalizing.**F1** : Train F1 gradually reaches 0.70+ whereas val F1 lags behind like accuracy.**AUROC** : The value jumps around 0.74 by 5th epoch. For the rest of the epochs it keeps oscillating between 0.76-0.79.
The model learns distinctive features in early epochs and then starts to overfit the training data.All four graphs confirm that.Early stopping helps to prevent further degradtation.

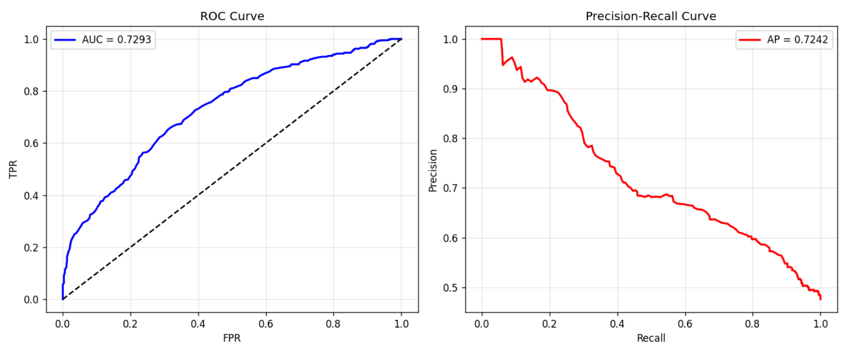

Fig. 7. ROC curve and Precision-Recall curve of the XceptionNet on the test set.

In Fig.7, the left plot shows the ROC curve with ACU = 0.7293.Here the curve bow is above the dashed diagonal line, which means the model can distinguish between classes,but it is still below the 0.9+ level, which is expected for model to be a good deepfake detector.It rises sharply in the beginning showing good sensetivity al low FPR.Later on it flattens indicating that model is confused. The right plot shows that in the beginning the precision is very high(1.0) when recall is low which means models is confident when it classifies data as fake. But as recall increases precision start to degrade and is at 0.5 when recall is 1.0 which means model is now flagging false positive as fake.

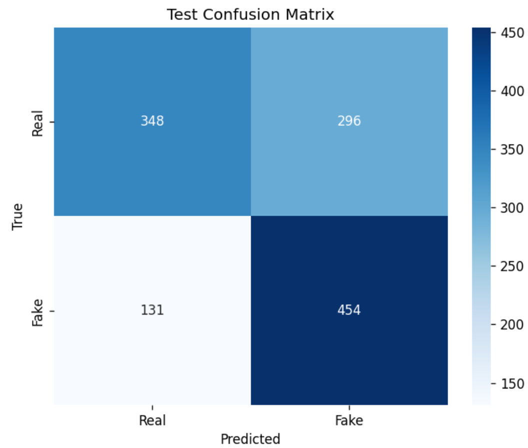

Fig. 8. Confusion matrix of XceptionNet model

In Fig.8 Confusion Matrix reveals that out of real samples 348 were correctly classified as real, 296 were incorrectly classified as fake.For fake samples 454 samples were correctly classified as fake and 131 were misclassified as real. This means model is biased towards fake class, indicating XceptionNet is effective at spotting deepfake but needs improvement for real videos.

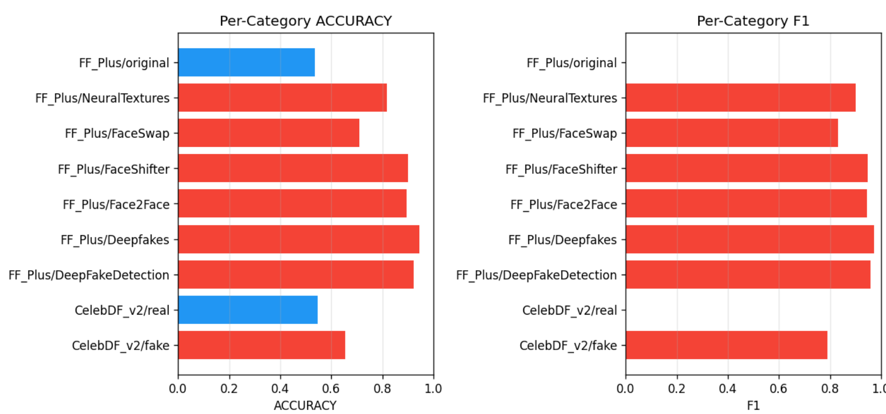

Fig. 9. Per-category performance of XceptionNet across FaceForensics++ and Celeb-DF v2 subsets

In Fig.9, We can clearly see that our model gives highest values for FF++ DeepFake category. Model fails for F++ original category and for Celeb-DF real category which means model is classifying almost everything as fake for original and real videos. This happens because fake-classes occupies a lager part of the feature space.

7 Discussion

The project tries to answer the question whether a model can learn to detect deepfakes by looking at physiological signal maps rather than pixel-level visual artifacts? The results show that the answer is yes but the difficulty of detection varies greatly on how capable the model is.

- Key Findings: The baseline CNN model achieved 59.2% accuracy and an AUC-ROC of 0.620 on the test set. This is better than random chance but only slightly so and the model struggled particularly to detect fake samples shown by F1 score as only 0.522 for the fake class. The XceptionNet model was pre-trained on ImageNet and then fine-tuned on our dataset showed substantially better results with 65.3% accuracy, AUC-ROC of 0.729, and an F1 score of 0.680 for fake detection. The two models showed a difference of roughly 6 percentage points in accuracy and 0.11 in AUC which is the main empirical finding of this project and it confirms that pretrained deep architectures bring meaningful benefits even when the input is PPG maps whicha re very different from usual videos or photo dataset.
- PPG maps encode physiological signal consistency across facial regions as a subtle spatially distributed pattern that requires understanding of relationships at multiple scales simultaneously. The CNN does not have the depth or feature-learning capacity to build this kind of understanding from scratch. The model training spent the first 11 epochs producing essentially random outputs (loss stuck near 0.693, accuracy at 50%) before adaptive learning rate reduction finally kicked in and meaningful learning began. A significant portion of the training budget had already been wasted by the time the model started converging.
- The per-category breakdown from the XceptionNet results indicate that the overall accuracy number as obscure. On FF++ manipulation types, XceptionNet performs remarkably well such as 94.5% on Deepfakes, 92.3% on DeepFakeDetection, 89.7% on Face2Face, 90.1% on FaceShifter, 81.8% on NeuralTextures, and 71.1% on FaceSwap. These are good numbers and suggest that the model has genuinely learned meaningful features specific to how each manipulation method disrupts physiological signals. However, the model performs poorly on the real class from both FF++ (53.6%) and CelebDF-v2 (54.6%). This pattern of high accuracy on FF++ fakes, low accuracy on real samples and CelebDF fakes points to a cross-dataset generalisation gap rather than a model failure. The model learned the FF++ fake distribution very well but did not generalise its understanding of "real" faces across both datasets equally.
- The most important comparison is with Ciftci et al.'s original FakeCatcher work, which is the direct inspiration for our project. Their 91.1% accuracy on FF++ uses a carefully engineered SVM pipeline built on top of rPPG features and is not end-to-end learnable. This project attempted to replace that pipeline with an end-to-end CNN that learns directly from the PPG maps and achieved 65.3% on a harder multi-dataset. The gap is significant but understandable as the original FakeCatcher method uses purpose-built physiological features that encode expert knowledge about how blood flow patterns behave, while

the CNN must discover these patterns from scratch. The comparison with Rossler et al.'s XceptionNet results which showed 96% accuracy on FF++is also instructive but less direct because their model works directly on RGB video frames and benefits from rich pixel-level deepfake artifacts that are much more visually obvious than the subtle signal differences captured in PPG maps. The fact that this project's XceptionNet achieves 65.3% on PPG maps rather than 96% on RGB frames demonstrates how much harder the physiological signal formulation of the problem is and not that the model is worse.

- The results of this project show an interesting intersection of computer vision and biometric signal processing. The core finding is that a pretrained vision model can extract features from physiological signal maps well enough to beat a CNN made from scratch by a meaningful and good margin. It also has a practical implication that matters for the field for example you do not need to build a bespoke physiological signal classifier from scratch to make PPG-based deepfake detection work.
- There are several honest limitations worth acknowledging. First, the test set was not completely balanced across manipulation types because during one of the run sessions, the Deepfakes and DeepFakeDetection folders were missing from the test partition due to an incomplete Drive sync, which reduced the fake count and created a slight real-class majority in the test split. This makes direct numerical comparison across different runs of the experiment less reliable than it should be.
  Second, the dataset itself has a known demographic skew. Both FF++ and CelebDF-v2 were constructed primarily with European subject meaning that neither the training distribution nor the evaluation reflects the full diversity of real-world faces. A model trained and evaluated on this data may generalise poorly to faces that are from other parts of world. Thus posing as an important fairness concern that this project was not designed to address.
  Third, both the models, Baseline CNN and XceptionNet were overfitting due to data availability and processing limit constraints.
- Future directions: One of the next step is completing the XceptionNet pipeline to generalize the model cross dataset and perform experiments where the model is trained on FF++ alone and tested on CelebDF-v2 alone and vice versa. This would give a much cleaner picture of how well the learned PPG features generalise across generation methods.
  Beyond this project's immediate scope, several other directions also look promising such as incorporating temporal information explicitly could allow the model to detect inconsistencies in the temporal evolution of physiological signals that are invisible in any single spatial map. A 3D CNN or a vision transformer applied to stacked PPG windows would be a natural architecture choice for this.

8 Conclusion

The project explores PPG map based deepFake detection using two models : Baseline CNN and XceptionNet. Where the model was trained two combined datasets : FaceForensics++ and Celeb-Df v2. The core idea is to generate PPG maps by extracting GPPG and CPPG signals, then taking difference of these two generate 3-channel input map of size 224x224x3. XceptionNet significantly performed better than baseline CNN in terms of every metric AUROC score was XceptionNet is 0.72 where as for CNN it is 0.62 .The pretrained ImageNet weigths gave XceptionNet a head start and depthwise separable convolution learned rich textures and edges. The model struggles with cross dataset generalization. The fake count is better because FF++ has 6 deepfake categories due to which model is learning deepfake patterns reals well. We were not able to resolve this overfitting and generalization issue beacause we were constrained by GPU limit and dataset diversity.

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